Prenatal Developmental Timing Is Process-Specific

The same prenatal exposure can have different effects depending on dose, route, and the developmental process underway; pregnancy is not a single window.

GRADE: MODERATE-PRINCIPLE; HIGH-NO-SINGLE-WINDOW | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Susceptibility to a prenatal exposure varies with developmental stage, dose, and the process underway. Organogenesis is more vulnerable to structural malformation than later fetal growth; the same agent is not one “prenatal effect.”

EVIDENCE AND CAUSAL ASSESSMENT

Wilson’s principles, still treated as foundational in later reviews (Ujházy et al. 2012), state that teratogenic susceptibility varies with developmental stage, that access to embryonic tissues matters, and that manifestations increase with dose from no-effect to lethal. Human application rests on clinical timing (e.g. organogenesis vs later growth/functional injury), not on randomized timed exposures.

EFFECT SIZE AND PERSISTENCE

There is no universal effect size. The unit is exposure × window × process × outcome. Persistence must be measured for each pairing; a sensitive period is not an inevitable-injury window.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (embryological organogenesis windows, not an observational population phenotype association).

Model-system translation; sparse direct human observation of early embryos; dating error; selection into ART and clinical series.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean one missed week “ruins” development, or that every prenatal difference is a malformation.

WHAT ACTION IS REASONABLY JUSTIFIED

Public-health timing for known teratogens and periconceptional folate is justified. Treating pregnancy as a single optimization window is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Ujházy et al. (2012), Interdisciplinary Toxicology (Wilson’s principles)